



P21

End-to-End Evidence for Million-Scale Coordination and Its Extrapolation Boundary: Why MAPF Success Does Not Travel to Semantic-Judgment Multi-Agent Systems

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End-to-End Evidence for Million-Scale Coordination and Its Extrapolation Boundary: Why MAPF Success Does Not Travel to Semantic-Judgment Multi-Agent Systems

Working Title (EN): End-to-End Evidence for Million-Scale Coordination and Its Extrapolation Boundary: Why MAPF Success Does Not Travel to Semantic-Judgment Multi-Agent Systems

Volume P21 • the end-to-end dialogue companion to P1 "Million-Scale Extrapolation of Coordination Complexity of Layered Hybrid Topologies" • paired with P22 "The Memory Wall Is Not the Coordination Wall" as the twin "scale extrapolation boundary" papers • UDOS Reasoning Engine v7.6.0 • Fang Wenxin • 2026-09-21

One sentence first: MAPF (multi-agent pathfinding) is currently one of the few engineering fields that have run "million-Agent coordination" to the magnitude of end-to-end success rate, but the premise of its success is **geometric state space, full observability, structured tasks**; the work-order system UDOS cares about (with hidden ground truth, semantic acceptance, Byzantine faults) differs essentially from MAPF—MAPF proves "million-scale coordination is feasible on highly structured tasks," **not** "multi-Agent systems can expand million-scale on semantic judgment." This paper nails down this extrapolation boundary.

Evidence caliber (stated only once in the full text): this paper is a theory/boundary-analysis paper containing no new UDOS measured numbers; the UDOS conclusions cited all return to v7.5.0 P1 and its reports7/*.json (explicit / analytical grading). External literature is verified per EVIDENCE_LEDGER_v76.md: the real title of MAPF-GPT-DDG is Advancing Learnable Multi-Agent Pathfinding Solvers with Active Fine-Tuning (arXiv:2506.23793, IROS 2025), and its **DDG active incremental fine-tuning is true**; but the circulated "2048×2048 map / 524,000 Agents 100% / 1,000,000 Agents 99.9% / 163 microseconds per Agent" is **not found in that paper's arXiv abstract, only in Russian-media retelling**, and this volume cites it at a downgraded caliber, not as paper fact. Its predecessor MAPF-GPT (arXiv:2409.00134, AAAI 2025) actually measured only small maps, 16–32 Agents.

Abstract

Can multi-Agent systems scale to the million level? MAPF-GPT-DDG (arXiv:2506.23793, IROS 2025) is the most-cited success case for this question in the physical-coordination field. This paper does two things. First, **calibrate the evidence caliber**: separate the part of MAPF-GPT-DDG that truly holds (Delta-Data-Generation active incremental fine-tuning, learnable MAPF solver surpassing contemporaries) from the circulated "million-scale 99.9%" rumor—the latter, verified per EVIDENCE_LEDGER, is found only in media retelling, and the paper's predecessor actually measured only 16–32 Agents, small maps. Second, **nail down the extrapolation boundary**: MAPF's success relies on three premises—geometricized state space (position/orientation are continuous coordinates), full task observability, structured acceptance criteria (collision-free is success); these three all fail in UDOS's semantic work-order system—work orders contain hidden ground truth, semantic acceptance is ambiguous, and Byzantine Agents lie. So MAPF's end-to-end success rate cannot serve as evidence that "semantic multi-Agent systems can scale million-scale." This paper simultaneously notes: the fan-in/rounds P1 measures at the protocol layer (star center 20.97 million messages vs layered tree constant 9, 14 rounds at the million tier) and the success rate MAPF measures at the task layer are **complementary rather than substitutable**—P1 proves the growth order of coordination overhead is controllable, MAPF (at its real caliber) proves

coordination on structured tasks is feasible; the "semantic-task end-to-end million-scale evidence" between the two is currently a blank. This paper gives three falsifiable conditions.

Keywords: multi-agent pathfinding; large-scale coordination; end-to-end success rate; extrapolation boundary; task structure; semantic acceptance; protocol layer vs task layer

Structured Abstract (Background problem → Argument → Evidence → Contributions)

- **Background problem:** external narratives often use "some system coordinated a million Agents" to prove multi-Agent is already scalable. But "coordinating a million Agents to complete A→B pathfinding" and "coordinating a million Agents to complete semantic acceptance with hidden ground truth" are two entirely different problems, and mixing them into one phrase "million scale" seriously misleads architecture decisions.
 - **Core argument:** MAPF's million-scale coordination success is the success of **structured geometric tasks**, and its evidence cannot be extrapolated to the **semantic-judgment multi-Agent systems** UDOS cares about; P1's protocol-layer complexity conclusion and MAPF's task-layer success rate are complementary, and neither alone can claim "semantic multi-Agent can be million-scale."
 - **Evidence threads:** (1) the truly holding part of MAPF-GPT-DDG (DDG active fine-tuning, arXiv: 2506.23793); (2) its predecessor MAPF-GPT actually measuring 16–32 Agents, small maps (arXiv: 2409.00134); (3) "million 99.9%" only media retelling, cited downgraded; (4) UDOS P1 protocol-layer explicit/analytical graded extrapolation (cpu-proto).
 - **Contributions:** splitting "million-scale coordination" into protocol/task/semantic three layers; explicitly listing MAPF's success premises and contrasting them with the UDOS work-order system; pointing out "semantic-task end-to-end million-scale evidence" is the current blank; three falsifiable conditions.
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1 Introduction: How Much Difference One Phrase "Million Scale" Hides

"We have coordinated a million Agents."—this phrase repeatedly appears in 2026 multi-Agent promotion. But when pressed "what are these million Agents doing," the answer often falls into two entirely different task classes: one is **physical coordination** (pathfinding, swarms, logistics scheduling), the other **semantic judgment** (code review, policy analysis, research reasoning). Putting these two task classes' "million-scale success" into one phrase is the most dangerous confusion of the current scale narrative.

UDOS P1 has already measured coordination complexity at the protocol layer: at a million nodes the star topology's center must handle about 20.97 million messages over about 8.39 million serial rounds, while the b-ary layered tree keeps every node's maximum fan-in constant at 9 and parallel rounds only 14 (million tier, cpu-proto, ≤4000-node explicit cross-check error 0, larger tiers analytical extrapolation). This is a **protocol-layer** conclusion—it says "how messages are organized without exploding" but not "whether things are done correctly after being organized."

MAPF-GPT-DDG is the most-cited "task-layer success" on this gap. What this paper does is neither erase its real contribution to physical coordination nor allow it to be extrapolated into evidence for semantic multi-Agents.

2 Evidence Calibration: What MAPF-GPT-DDG Really Establishes

2.1 The Truly Holding Part

Verified per EVIDENCE_LEDGER_v76.md (2026-09-21), MAPF-GPT-DDG really exists, its real title Advancing Learnable Multi-Agent Pathfinding Solvers with Active Fine-Tuning (arXiv:2506.23793, IROS 2025). Its truly holding contribution is **Delta-Data-Generation (DDG) active incremental fine-tuning**: not passively collecting training data but actively generating "informative incremental data" to fine-tune the MAPF-GPT solver, making it surpass all contemporary learnable solvers. This idea of "actively generating informative data" is isomorphic to UDOS P4's coverage-aware active collection and P18's breadth-depth two-stage flywheel—all "collecting data into blind zones."

2.2 The "Million Numbers" That Must Be Downgraded

A widely circulated set of numbers—"on a 2048×2048 map coordinating 524,000 Agents at 100% success, 1,000,000 Agents at 99.9%, decision speed 163 microseconds/Agent"—verified per this volume is **not found in that paper's arXiv abstract**, and its predecessor MAPF-GPT (arXiv:2409.00134, AAAI 2025) actually used small 17×17 to 21×21 maps, 16–32 Agents. This set of "million" numbers comes from a Russian-media retelling, and **this volume does not accept it as paper fact**.

So this paper's citation of MAPF is strictly limited to: DDG active fine-tuning is true, MAPF-GPT being small-map small-scale evidence is true; "million-scale 99.9%" is mentioned only as a **media claim**, labeled "not verified in the paper." This is the same discipline as UDOS P1's honest labeling of its own million-scale extrapolation (explicit / analytical grading).

2.3 Three Premises of MAPF Success

To understand why MAPF's success cannot be extrapolated, one must first see on what soil it succeeds.

Table 1 Three premises of MAPF success vs the UDOS work-order system

Premise	MAPF (pathfinding)	UDOS work-order system
State space	geometric: position/orientation are continuous coordinates, the constraint is "no collision"	semantic: tasks contain hidden ground truth, acceptance ambiguous
Observability	fully observable: map and other vehicles' positions known	partially observable: experts each hold private information
Acceptance criteria	structured: collision-free arrival is success, mechanically decidable	semantic: acceptance needs LLM/human judgment, with Byzantine false reports

These three contrast rows are this paper's core: MAPF reduces "coordination success" into a **mechanically decidable geometric-constraint satisfaction problem**; UDOS work orders leave "coordination success" to **ambiguous, possibly maliciously manipulated semantic judgment**.

3 Protocol Layer vs Task Layer: P1 and MAPF Are Complementary, Not Substitutable

Figure 1 Three layers of scale extrapolation: protocol / task / semantic

Figure 1 "Million-scale coordination" is in fact three different propositions: the protocol layer (message organization without exploding), the task layer (structured geometric-task success rate), the semantic

layer (semantic acceptance with hidden ground truth). P1 proves the first, MAPF at its real caliber touches the second, the third is currently blank. Conceptual illustration, not measured data.

P1 measures the **protocol layer**: in a deterministic discrete-event simulator with zero network and zero large-model calls, star center fan-in grows linearly with N (million tier about 20.97 million messages, about 8.39 million serial rounds), while the b -ary layered tree ($b=8$) keeps every node's maximum fan-in constant at $b+1=9$ and parallel rounds at $2\log_b N$ (million tier 14 rounds, cpu-proto). This proves "**how messages are organized without exploding with N** ."

MAPF at its real caliber measures the **task layer**: on a geometric map, whether Agents can reach goals collision-free. This proves "**coordination on structured tasks is engineering-feasible**."

There is a clear division between the two: P1 does not touch "whether done correctly," only "whether messages explode"; MAPF does not touch "whether acceptance is ambiguous," only "whether there is collision." And the **semantic layer** UDOS truly cares about—coordinating a million Agents to do tasks with hidden ground truth, needing semantic acceptance, possibly meeting Byzantine false reports—currently **has neither P1-style protocol-layer evidence (P1 contains no semantic faults) nor MAPF-style task-layer evidence (MAPF contains no semantic acceptance)**. This is the current blank and also the position v7.6.0 volume P3/P26/P31 is to fill.

4 Why Extrapolation Is Dangerous: Replacing Geometric Constraints with Semantic Acceptance

4.1 MAPF's "Success" Is Constraint Satisfaction, Not Correct Judgment

In MAPF "coordination success" has a precise definition: all Agents reach goals collision-free. This definition can be mechanically checked by a program—no LLM is needed to "judge" whether this step is correct. Precisely because acceptance is mechanical, the failure modes of large-scale coordination concentrate at the geometric level (deadlock, collision, local optimum), solvable by smarter pathfinding algorithms.

In the UDOS work-order system "coordination success" has no such mechanical definition. Whether a specialist's solution is correct depends on hidden ground truth (P6 has honestly recorded the identifiability blind zone where the acceleration scalar skill is negative); whether a QA's acceptance is honest depends on whether it is a Byzantine node (P2/P26). Replacing "collision-free" with "semantically correct" and "map known" with "ground truth hidden" changes the nature of the coordination problem.

4.2 The Role of Scale Differs in the Two Task Classes

In MAPF, increasing the Agent count mainly increases **geometric congestion**—more people more easily jam, a physical problem mitigable by better pathfinding algorithms. In semantic work orders, increasing the Agent count mainly increases **judgment ambiguity and the Byzantine surface**—with more people, whose word counts, who is lying, who is dragging, these governance problems are the bottleneck (P9 has proved homogeneous copies bring no quality improvement, P17 has found permission granting is a management bottleneck).

So "MAPF million-scale success" constitutes **no positive evidence** for the scalability of semantic multi-Agent systems. It at most shows: where the semantic problem is reduced to a geometric problem, million-scale coordination is engineerable.

It is worth separately noting the positive value of MAPF-GPT-DDG's DDG idea to UDOS, lest the "extrapolation boundary" be misread as "worthless." DDG's core is not "running a lot" but "actively generating informative incremental data"—not passively waiting for data but actively collecting into blind zones. This is isomorphic to UDOS P4's coverage-aware active collection and P18's breadth-depth two-stage flywheel: all "knowing where is not yet covered and specifically filling there." The only difference is DDG fills the geometric solver's training samples while UDOS fills governance and acceptance boundary samples. This methodology can be borrowed with confidence, and when borrowing one need not borrow its set of unverified million numbers.

4.3 Echo with P22

P22 will point out that Warp-Cortex-like architectures press memory complexity from $O(N \cdot L)$ to $O(1) + O(N \cdot k)$, but what it solves is the **memory wall**, not the coordination wall. MAPF solves the **geometric coordination wall**, not the semantic coordination wall. The two together convey the same boundary: the 2026 "million-scale" breakthroughs, taken apart, respectively tore down the memory wall and the geometric coordination wall, but the **semantic coordination wall** remains.

5 Dross, Failure Boundaries, and Falsifiable Conditions

5.1 Dross and circulated exaggeration

1. **Taking media retelling as paper fact.** The set "524k/100%/1M/99.9%/163μs" is the most directly cited yet least verifiable part of the MAPF narrative. Writing it as "MAPF-GPT-DDG proves million-scale 99.9%" is a violation of evidence discipline—this volume has downgraded it to a media claim.
2. **Generalizing "structured-task success" into "multi-Agent system success."** MAPF's Agents are homogeneous, tasks geometric, acceptance mechanical; UDOS's Agents are heterogeneous, tasks semantic, acceptance ambiguous. Extrapolating across these three differences is the core fallacy this paper opposes.
3. **Taking "end-to-end success rate" as sufficient evidence for "scalable."** N Agents completing a task in one demo does not equal still scalable under fault injection, partial observability, semantic ambiguity. The reason P1 insists on explicit/analytical grading is precisely to reject this demo-style extrapolation.

5.2 Failure boundaries

- This paper neither denies MAPF's value in the physical-coordination field nor the DDG active fine-tuning idea; this paper only draws clear its evidence applicability to **semantic multi-Agent systems**.
- If in the future there appears end-to-end million-scale coordination evidence for a semantic task (such as long-horizon code projects, complex policy analysis), still stably completing under fault injection and Byzantine false reports, then this paper's "semantic-layer blank" judgment needs upward revision.

5.3 Falsifiable conditions

- **C1:** if it can be proved under the same caliber that the large-scale success rate of MAPF-style geometric coordination remains unchanged after introducing semantic acceptance, hidden ground truth, and Byzantine nodes, then the "extrapolation boundary" is weakened—but this needs a hybrid benchmark containing both geometry and semantics, currently nonexistent.

- **C2:** if the body of the MAPF-GPT-DDG original paper (arXiv:2506.23793) does contain the "million-scale 99.9%" measurement, then this paper's downgrade of that number needs withdrawal; conversely if final verification confirms that number only comes from media retelling, then this volume's caliber holds.
 - **C3:** if UDOS P1's protocol-layer extrapolation (layered-tree fan-in constant 9, million-tier 14 rounds) is proved in real distributed deployment to degrade greatly due to communication delay or message loss, then the gap between "protocol-layer controllable" and "task-layer success" is larger than this paper estimates.
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6 Hard Predictions (2026–2031, post-hoc scorable)

- **F1:** within the next five years, there will not appear an independently reproduced million-scale end-to-end success-rate evidence on a **semantic-judgment** multi-Agent system (with hidden ground truth, semantic acceptance, Byzantine faults); wherever a "million-scale" narrative appears, taken apart one finds its task still biased toward structured (geometry/scheduling/retrieval) or its scale numbers come from media rather than papers.
 - **F2:** the first true "semantic-layer million-scale coordination" evidence will come not from pathfinding or social simulation but from long-horizon software engineering or complex scientific reasoning—i.e. the task itself must first possess mechanically decidable acceptance (such as tests passing, paper reproduction), otherwise the governance bottleneck erupts before the scale bottleneck.
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7 Conclusion

MAPF-GPT-DDG is a real contribution in the physical-coordination field: its DDG active fine-tuning idea is worth UDOS P4/P18 borrowing. But the circulated "million-scale 99.9%" does not stand verification, and even if verified, what it proves is only "million-scale coordination on structured geometric tasks is feasible," not "semantic multi-Agent systems can scale million-scale."

The relationship of P1 and MAPF should be placed thus: P1 at the protocol layer proves the growth order of message organization is controllable (cpu-proto), MAPF at the task layer (at its real caliber) proves structured geometric coordination is engineerable; the two are complementary, while **the semantic-layer end-to-end million-scale evidence remains blank**. For engineering, this means before claiming "our multi-Agent system can be million-scale," one must first answer: are these million Agents doing geometric or semantic tasks? Is acceptance mechanical or ambiguous? Does it complete when meeting Byzantine nodes? If any one of the three questions is unanswered, "million scale" is only a scale narrative.

Million-scale coordination is not one proposition but three. Saying them mixed together is disguising engineering ambition as engineering fact.

References

Classics and background

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UDOS paper volume (this series, evidence grades marked in text)

1. UDOS v7.5.0: P1 "Million-Scale Extrapolation of Coordination Complexity of Layered Hybrid Topologies", P6 "Observability and Identifiability of Hidden Dynamic Parameters", P9 "The Quality Saturation Law of Agent Legions".
 2. UDOS v7.6.0: P20 "Heterogeneity Dividend and Homogeneity Saturation", P22 "The Memory Wall Is Not the Coordination Wall", P26 "Weighted BFT and Reputation".
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Evidence Discipline and Reproduction Notes

- This paper is a theory/boundary-analysis paper with no new experiments; all UDOS numbers return to P1 and `reports7/`, evidence grades per P1's labeling (≤ 4000 nodes explicit, larger tiers analytical, single-seed cpu-proto pilot).
- MAPF-GPT-DDG's "million-scale 99.9% / 163 μ s" is downgraded per EVIDENCE_LEDGER_v76.md item A8 to **media retelling • not verified in the paper**, and this paper does not write it as paper fact; DDG active fine-tuning and the MAPF-GPT predecessor's small-map measurement are verified facts.
- This paper's predictions (F1–F2) and falsification conditions (C1–C3) can be post-hoc scored with public evidence during 2026–2031.